

The Right Kind of Volatility

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Uncertainty Zones and ticks

- UZ had size $2 \cdot \eta \cdot \alpha$ and is centered at mid-ticks ($P_i + \frac{\alpha}{2}$)

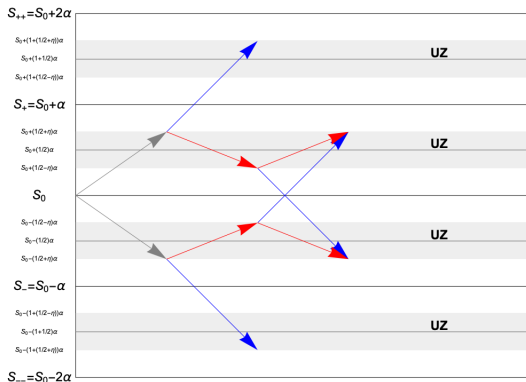


Figure: Uncertainty Zones Points; alternations in red, continuations in blue

Signed alternations and continuations

- Defining the 4 kinds of moves

			Next efficient price	
			ud = +1	ud = -1
Last trade sign		Last efficient price	$S_0 + \left(\frac{1}{2} + \eta\right) \cdot \alpha$	$S_0 - \left(\frac{1}{2} + \eta\right) \cdot \alpha$
Down	sud = +1	$S_0 + \left(\frac{1}{2} - \eta\right) \cdot \alpha$	Alt Up	Cont Down
Up	sud = -1	$S_0 - \left(\frac{1}{2} - \eta\right) \cdot \alpha$	Cont Up	Alt Down

Estimation in the UZ model

The efficient price process X_t is described by:

$$\frac{dX_t}{X_t} = \mu_t dt + \sigma_t dW_t$$

$$\omega_t = \mu_t - \frac{1}{2}\sigma_t^2$$

For the $\mu = 0$ case we have (RR2009):

$$\hat{\eta} = \sum_k (\lambda_k \cdot u_k)$$

$$\lambda_k = \frac{N_k^A + N_k^C}{\sum_k (N_k^A + N_k^C)}$$

and

$$u_k = \frac{k \cdot N_k^C}{2 \cdot N_k^A} - \left(\frac{k-1}{2} \right)$$

Conditional CDFs

- Conditional durations (Δt to next price change) are described by (the sign $\pm$ refers to ud):

$$P_{\pm}(t) = +e^{\frac{\omega}{\sigma^2} \cdot (b_1 - x_0)} \cdot \sum_{n=1}^{\infty} \left\{ e^{\frac{\omega}{\sigma^2} \cdot y_n^{\pm}} \cdot N\left(\frac{y_n^{\pm} + \omega \cdot t}{\sigma \cdot \sqrt{t}}\right) + e^{-\frac{\omega}{\sigma^2} \cdot y_n^{\pm}} \cdot N\left(\frac{y_n^{\pm} - \omega \cdot t}{\sigma \cdot \sqrt{t}}\right) \right\} \\ - e^{\frac{\omega}{\sigma^2} \cdot (b_1 - x_0)} \cdot \sum_{n=1}^{\infty} \left\{ e^{\frac{\omega}{\sigma^2} \cdot z_n^{\pm}} \cdot N\left(\frac{z_n^{\pm} + \omega \cdot t}{\sigma \cdot \sqrt{t}}\right) + e^{-\frac{\omega}{\sigma^2} \cdot z_n^{\pm}} \cdot N\left(\frac{z_n^{\pm} - \omega \cdot t}{\sigma \cdot \sqrt{t}}\right) \right\}$$

- $b_1 = \text{Log}[S_0 + ud \cdot (\frac{1}{2} + \eta) \cdot \alpha]$
- $b_2 = \text{Log}[S_0 - ud \cdot (\frac{1}{2} + \eta) \cdot \alpha]$
- $x_0 = \text{Log}[S_0 + sud \cdot (\frac{1}{2} + \eta) \cdot \alpha]$
- $y_1 = +ud \cdot (x_0 - b_1) < 0$
- $y_n = y_1 + ud \cdot 2 \cdot (n - 1) \cdot (b_2 - b_1) < y_1$
- $z_n = y_n + ud \cdot 2 \cdot (b_2 - x_0) < y_n$

To infinity ... and beyond

- The limits for $t \rightarrow \infty$ are:

$$P_{\pm}(\omega \neq 0) = e^{\left(\frac{1 - \text{sign}[\omega] \cdot ud}{2}\right) \cdot (b_1 - x_0)} \cdot \left(\frac{1 - e^{\left(\frac{2 \cdot \text{abs}[\omega] \cdot ud}{\sigma^2}\right) \cdot (b_2 - x_0)}}{1 - e^{\left(\frac{2 \cdot \text{abs}[\omega] \cdot ud}{\sigma^2}\right) \cdot (b_2 - b_1)}} \right)$$

$$P_{\pm}(\omega = 0) = \frac{b_2 - x_0}{b_2 - b_1}$$

- So given the sign of the last trade: $sud_j = -ud_{j-1}$
- We know $S_0 + sud \left(\frac{1}{2} - \eta\right) \cdot \alpha$ and the probabilities of an up or down move
- And based on which move happens we know the CDF of the durations

Logistic approximation

- The formulas from the previous slide for continuations can be approximated by:

$$P_{\pm}^C = \left(\frac{2 \cdot \eta}{1 + 2 \cdot \eta} \right) \cdot \left(\frac{2}{1 + e^{\left(\mp 2 \cdot \frac{\mu}{\sigma^2} \cdot \frac{\alpha}{S_0} \right)}} \right)$$

$$P_{\mp}^A = 1 - P_{\pm}^C$$

- And average durations can be estimated by defining a scale τ :

$$\tau = 2 \cdot \eta \cdot \left(\frac{\alpha}{\sigma \cdot S} \right)^2$$

- For the $\mu = 0$ case the average conditional durations are:

$$\tau^A = \frac{2 + 2 \cdot \eta}{3} \cdot \tau$$

$$\tau^C = \frac{1 + 4 \cdot \eta}{6 \cdot \eta} \cdot \tau$$

The trend is not a friend

- For the $\mu \neq 0$ case the average duration gets modified (reduced) by a factor that is close to:

$$adj_{\tau} = \frac{1}{1 + \left(\frac{\mu}{2\sqrt{2} \cdot \sigma}\right)^2 \cdot \left(\frac{\alpha}{\sigma \cdot S}\right)^2}$$

- Later when we use the distribution of the durations for inference we will see that we benefit from looking at the cases where $\mu \neq 0$, as $\frac{\partial(\tau \cdot adj_{\tau})}{\partial \mu} = 0$ at $\mu = 0$

Large moves

- Continuations of k ticks:

$$P(k, \alpha, \eta, t) \rightarrow P\left(1, k \cdot \alpha, \frac{\eta}{k}, t\right)$$

- Alternations of k ticks:

$$P(k, \alpha, \eta, t) \rightarrow P\left(1, \alpha, \eta + \left(\frac{k-1}{2}\right), t\right)$$

- This is a really good way of dealing with these moves later with the bayesian estimator

Updated estimator

- We can write:

$$\hat{\eta} = \frac{\sum_k (k \cdot N_k^C)}{2 \cdot \sum_k (N_k^A)} - \sum_k \left(\lambda_k \cdot \left(\frac{k-1}{2} \right) \right)$$

- With

$$\lambda_k = \frac{N_k^A}{\sum_k (N_k^A)}$$

- Or directly:

$$\hat{\eta} = \frac{\sum_k (k \cdot N_k^C) - \sum_k ((k-1) \cdot N_k^A)}{2 \cdot \sum_k (N_k^A)}$$

“Markov” chain

- Adding the count of the 4 different events we have a “Markov” chain (events are not evenly spaced, probabilities may vary over time):

	Up	Down	Alt Up	Alt Down	Cont Up	Cont Down
Up				P_{-}^A	P_{+}^C	
Down			P_{+}^A			P_{-}^C
Alt Up	1					
Alt Down		1				
Cont Up	1					
Cont Down		1				

Asymptotic probabilities

- For Up and Down moves: $\rho_+ = \frac{P_+^A}{P_+^A + P_-^A}$ and $\rho_- = \frac{P_-^A}{P_+^A + P_-^A}$
- For Alternations (this holds well in the data I've seen so far):

$$\rho_+^A = \rho_-^A = \frac{P_+^A \cdot P_-^A}{P_+^A + P_-^A}$$

- For Continuations Up:

$$\rho_+^C = \frac{P_+^A \cdot (1 - P_-^A)}{P_+^A + P_-^A}$$

- For Continuations Down:

$$\rho_-^C = \frac{(1 - P_+^A) \cdot P_-^A}{P_+^A + P_-^A}$$

Simulation

- Each price change is determined by a pair of random numbers $\{r_1, r_2\}$:
 - The first determines whether the next trade is Up or Down given the sign of the last trade: $sud_j = -ud_{j-1}$
 - The second determines the duration between price changes:

$$\frac{P_{\pm}(t)}{P_{\pm}} = r_2$$

- The distribution of price changes of more than one tick can be a variable of the (parameters of the) simulation as well
- Not only realistic price changes but realistic durations

Summary of updated UZ model

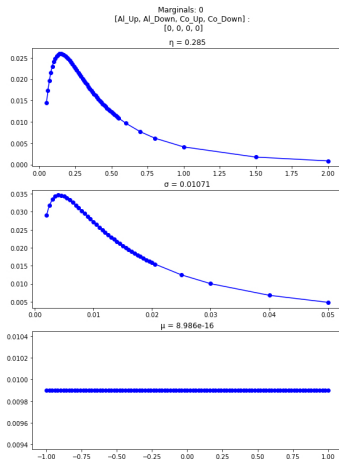
- Updated UZ model for a more general case ($\mu \neq 0$)
- Updated estimator for a better treatment of large moves
- Durations (unconditional and conditional) estimators including ($\mu \neq 0$)
- The count $\frac{N^C}{2 \cdot N^A}$ is now equal to $\frac{P_+^A + P_-^A}{4 \cdot P_+^A \cdot P_-^A} - \frac{1}{2} \geq \eta$
- Signed alternations and continuations with their own probabilities (and $\rho_+^A = \rho_-^A$)
- We will use the approximation $\rho_+^C - \rho_-^C = 2 \cdot \eta \cdot \tanh\left(\frac{\mu}{\sigma^2} \cdot \frac{\alpha}{S_0}\right)$ later
- Easy to simulate

Goal of BUZ

- Two different goals: (i) Historical / EOD and (ii) Online (our focus)
- Get a better estimate of intraday changes in $\theta = \{\eta, \sigma, \mu\}$
- Captures intraday change in relative tick size (as the price of the asset changes)
- No need for a rolling window to estimate η and σ : react quickly to changes
- Easy to calculate (trades: price changes only)
- Efficient prices estimates (σ and μ) can be combined across different contracts / assets with the same underlying

Priors and independence

- Start with lognormal priors for η and σ , uniform priors for μ and assume independence (easier)



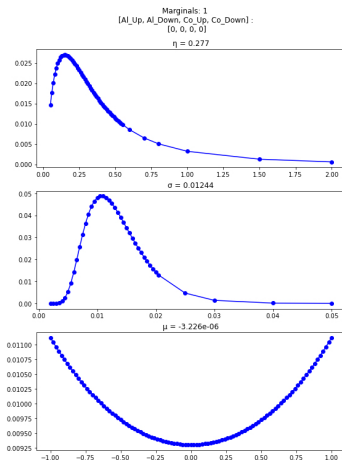
Durations

$$\theta = \{\eta, \sigma, \mu\} : p(\theta | x) = p(x | \theta) \cdot p(\theta)$$

- Where x is the relevant data for each change in price (tick size, sign and size of the move in ticks, time from previous price change)
- The likelihood $p(x | \theta)$ is given by the pdf corresponding to $P_{\pm}(t)$ as previously defined
- But remember that it won't help much with μ specially around $\mu = 0$
- In this particular implementation I discretized the grid quite coarsely
- Different paths for efficient price, same random numbers, parameters change midway through (could do it for tick size as well)
- Base case: $\theta = \{\eta, \sigma, \mu\} = \{0.35, 0.01, 0\}$

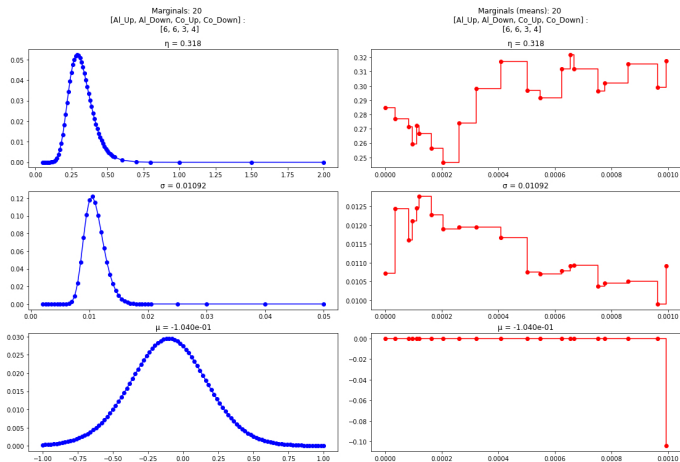
Give me a “t”

- Even the first price change is informative (it gives a good idea of the scale τ)



20 questions

- After 20 price changes things look better:



Counts

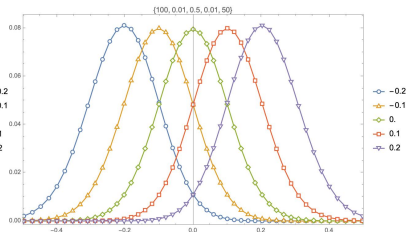
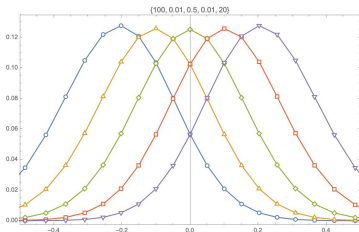
- How did we update μ after 20 price changes?
- The likelihood $p(x | \theta)$ is given by the multinomial likelihood:
 - $p(\{n_+^A, n_+^A, n_+^C, n_+^C\} | \{\rho_+^A, \rho_+^A, \rho_+^C, \rho_+^C\}) =$

$$\frac{(n_+^A + n_+^A + n_+^C + n_+^C)!}{(n_+^A! \cdot n_+^A! \cdot n_+^C! \cdot n_+^C!)} \left((\rho_+^A)^{n_+^A} \cdot (\rho_+^A)^{n_+^A} \cdot (\rho_+^C)^{n_+^C} \cdot (\rho_+^C)^{n_+^C} \right)$$
- Two different goals: (i) Historical / EOD and (ii) Online (our focus)
- For Historical we can get a large window (e.g. $n=100$) and use the posterior for μ back to recalculate θ (until it doesn't change)
- For Online we try to choose n that is big enough to give us some differentiation but not that big that will take too long to update or that will give us a likelihood too narrow
- We can use $n=20$ up to $n=50$; with $n=20$ we're bound to react too quickly, but updating the probabilities every 20 changes as well should be enough to dissipate transients quite quickly

Counts - Likelihoods

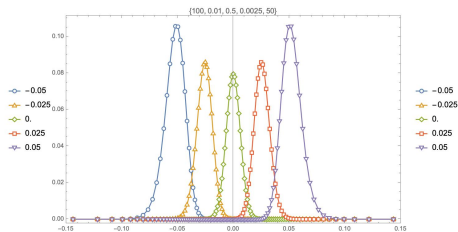
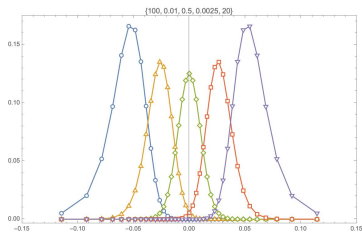
- From $\rho_+^C - \rho_-^C = 2 \cdot \eta \cdot \tanh\left(\frac{\mu}{\sigma^2} \cdot \frac{\alpha}{S_0}\right)$ we infer

$$\hat{\mu} = \frac{\hat{\sigma}^2}{\left(\frac{\alpha}{S_0}\right)} \cdot \tanh^{-1}\left(\frac{\rho_+^C - \rho_-^C}{2 \cdot \hat{\eta}}\right)$$



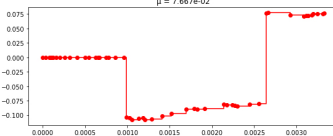
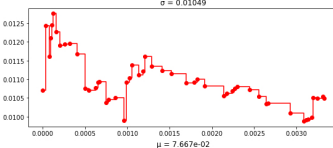
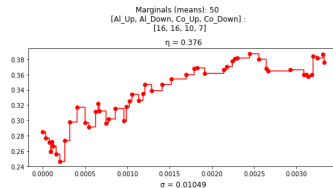
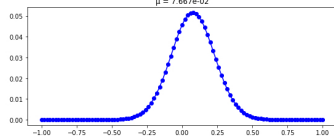
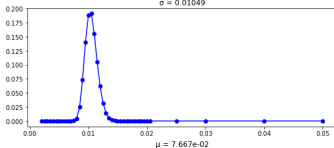
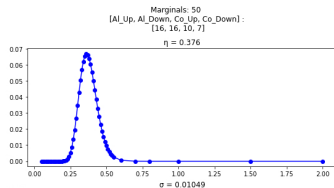
Counts - Likelihoods - Low σ

- A lower σ gives us better resolution for μ (but worse for η)



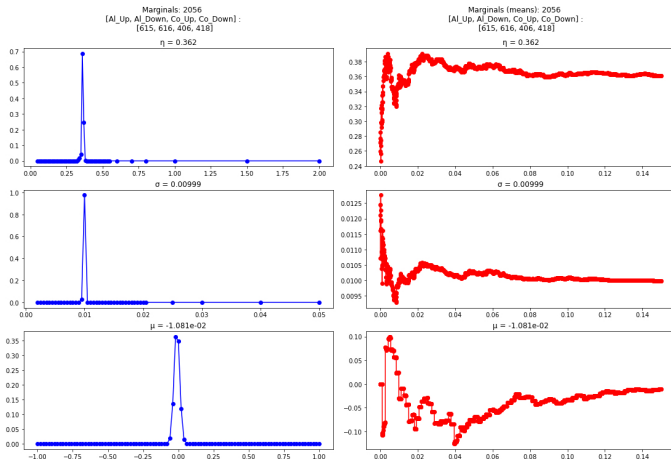
Are we there yet?

- After 50 price changes things look really good:



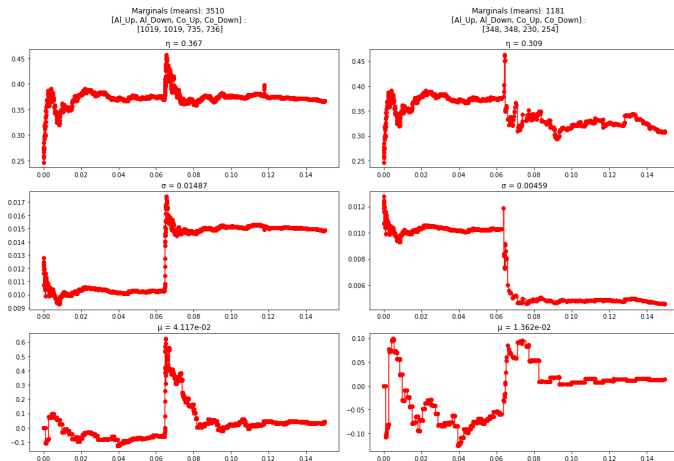
No surprises

- After 15% of the simulated trading day (9 hours long) our posteriors have become quite narrow and the estimates for η and σ are stable:



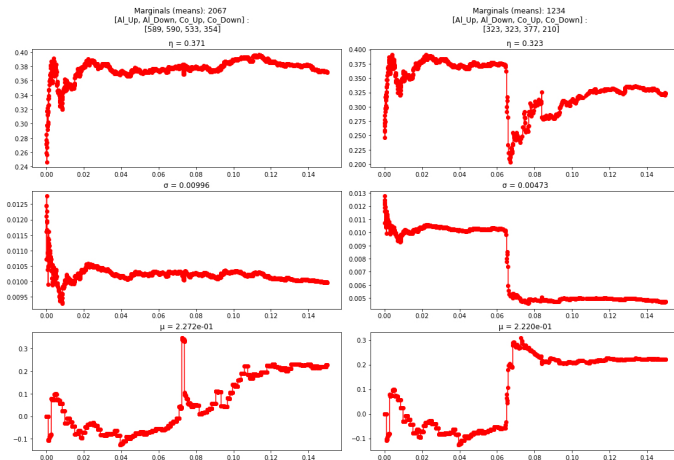
σ up and down

- We capture quickly changes in σ (from 0.01 to 0.015 on the left and 0.005 on the right), but increases make μ unstable and decreases make η unstable



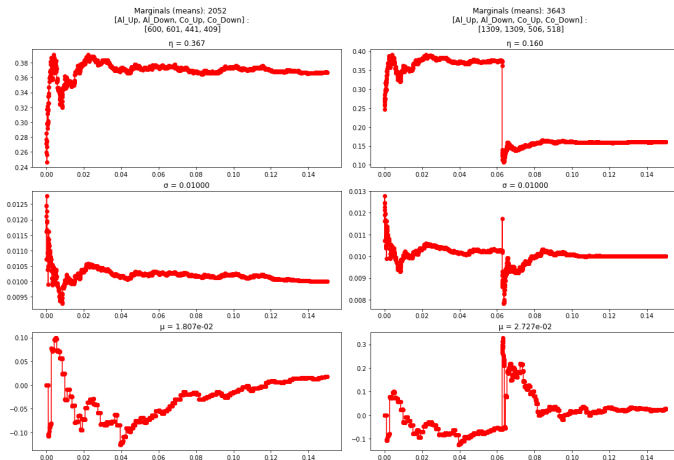
μ with different σ

- The higher the ratio μ/σ^2 , the better we are in estimating μ (going from 0 to 0.20)



Low μ ; change in η

- The lower the ratio μ/σ^2 , the harder it is to find μ (from 0 to 0.05); lower η (from 0.35 to 0.15) also make μ harder to estimate



The Uncertainty Principle of the Uncertainty Zones

- We cannot have good estimates of all 3 parameters η , σ and μ simultaneously
- ① High σ helps with η but worsens μ
- ② Low σ helps with μ but worsens η
- ③ High η helps with μ but worsens σ
- ④ Low η helps with σ but worsens μ
- ⑤ High μ helps with σ but worsens η
- ⑥ Low μ helps with η but worsens σ

Summary of Bayesian UZ estimation

- 1 Quick to find σ in most cases
- 2 Nudging μ away from 0 helps, but don't try be overconfident on μ
- 3 Cannot be excellent in all 3 simultaneously
- 4 n between 20 and 50 for counts should be good enough for online
- 5 Combine estimation (σ and μ) across different contracts with the same underlying

What determines η ?

- Probabilistic mean reversion parameter (can be higher than $1/2$)
- Increases with the dispersion of the distribution of the size of market orders
- Decreases with the increase of the posted amount on the top of the queue
- Decreases with the increase of the probability of a reverting fill of empty places on the top of the queue (reverting fill: keep the previous bid/ask status before cancel/trade)

Instability and feedbacks

- Agents are always trying to estimate the probabilities of reversion and trending; slow agents rely mainly on price changes, not order book dynamics

- What happens when the formulas:

$$\rho_+^C - \rho_-^C = 2 \cdot \eta \cdot \tanh \left(\frac{\mu}{\sigma^2} \cdot \frac{\alpha}{S_0} \right)$$

- And:

$$\hat{\mu} = \frac{\hat{\sigma}^2}{\left(\frac{\alpha}{S_0} \right)} \cdot \tanh^{-1} \left(\frac{\rho_+^C - \rho_-^C}{2 \cdot \hat{\eta}} \right)$$

- Feed on each other? And when σ increases (probably increasing η) and makes it harder to estimate μ ?

Where will it end?

- Look at stability (or lack of) in the matrix:

	Up	Down	Alt Up	Alt Down	Cont Up	Cont Down
Up				P_{-}^A	P_{+}^C	
Down			P_{+}^A			P_{-}^C
Alt Up	1					
Alt Down		1				
Cont Up	1					
Cont Down		1				

Stop the press

- What can exchanges and regulators do?
 - Standard answer: auctions after big (and/or fast) price changes
 - Alternative: Increase the tick size
 - Each limit order has an original price but it could have a “rounded” executable price depending on the prevailing tick size
 - You keep price discovery while keeping up with increasing volatility
 - Intraday changes in tick today are only dependent on price - why not volatility?

Rough volatility

- Enough about Monte Carlo paths, what about real data?
- Volatility is rough, so who's responsible: σ or μ ?
 - “Drift Burst”: once a week
 - So probably σ , but why and how?
- Hawkes process: reactive moves
- Consequence of electronic trading

“Trading: Fast and Slow”

- Mixture of two different processes on very different scales
- Classify durations
 - Fast should be of the order of magnitude of latencies (we should be able to see this over time) -
 - Slow should behave as above
- Then check if the “slow” volatility still fluctuates

Methods and code

- Improve grid (adaptive, more granularity around means as marginals narrow)
- Adaptive window for count as volatility changes
- Parallelization / memoization of likelihood calculations
- Parallel “fast” and “slow” processes
- Parallel σ and μ marginals for related assets

Data and models

- Futures with normal and mini versions (different η and sometimes α , low latency)
- Market shocks (drift burst?)
- Can we separate the “fast” and “slow” regimes?
- Is the behavior of large moves consistent?
- Link to Queue Reactive model
- Even price level changes are a natural experiment

What this talk was about anyway?

Volatility estimation

- Take a great model (UZ) and improve it (trends and large moves)
- Use durations and counts for bayesian estimation of volatility and trend

Market microstructure

- Generalize η as a mean reversion parameter (can be higher than 1/2)
- Quantify uncertainty as a driver of market-maker behavior
- Different time scales and possibly different behaviors



Books, papers

- Robert, C. Y. and Rosenbaum, M. (2009): “Volatility Estimation under Endogenous Microstructure Noise”
- Dayri, K. and Rosenbaum, M. (2015), “Large Tick Assets: Implicit Spread and Optimal Tick Size
- Huang, W., Lehalle, C.-A. and Rosenbaum, M. (2015), “Simulating and analyzing order book data: The queue-reactive model
- Carreira, M., and Brostowicz, R. (2016): “Brazilian Derivatives and Securities”
- Sidenius, J. (1998), “Double barrier options: valuation by path counting”
- Christensen, K., Oomen, R. and Renò, R. (2016): "The Drift Burst Hypothesis"
- Chaboud, A., Dao, A. and Vega, C. (2019): “What makes HFTs tick?”